SABR model validation

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1 Introduction

Option pricing models play an important role in pricing, hedging, and risk management. The Black model gives a simple baseline model for European option pricing. However, the assumption of constant volatility cannot explain the implied volatility smile observed in market data.

In this project, we study the stochastic alpha beta rho model, or SABR model, introduced by Hagan et al.[1] The main idea is to keep the computational convenience of the Black formula, while replacing the constant volatility by an asymptotic implied volatility generated from a stochastic volatility model.

Our goal is to implement the Black model and the SABR model, calibrate SABR parameters to market option data, compare model prices, and study hedge error.

2 Models

2.1 Black model

Hagan starts from writing down the call and put prices for the Black model. For a European call option,

$$\mathcal{C}_0 = D(t_{\text{set}})E\left[\left(\hat{F}(t_{\text{ex}}) - K\right)^+ \middle| \mathfrak{F}_0\right], \tag{1}$$

where $D(t_{\text{set}})$ is the discount factor to settlement, $\hat{F}(t)$ is the forward price process, t_{ex} is the expiration time, K is the strike, and $\mathfrak{F}_0$ is the initial information.

Under the forward measure, $\hat{F}(t)$ is a martingale. We write

$$d\hat{F} = C(t, \hat{F})dW, \qquad \hat{F}(0) = f, \tag{2}$$

where C is some coefficient depending on t and $\hat{F}$. For the Black model,

$$C(t, \hat{F}) = \sigma_B \hat{F}. \tag{3}$$

Thus

$$d\hat{F} = \sigma_B \hat{F} dW. \tag{4}$$

The closed form call and put prices can be derived by writing the expectation in terms of the integral over density functions:

$$\mathcal{C}_0 = D\left(f\mathcal{N}(d_1) - K\mathcal{N}(d_2)\right), \tag{5}$$
$$\mathcal{P}_0 = D\left(K\mathcal{N}(-d_2) - f\mathcal{N}(-d_1)\right), \tag{6}$$

where

$$d_1 = \frac{\log(f/K) + \frac{1}{2}\sigma_B^2 t_{\text{ex}}}{\sigma_B\sqrt{t_{\text{ex}}}}, \tag{7}$$
$$d_2 = d_1 - \sigma_B\sqrt{t_{\text{ex}}}. \tag{8}$$

The call delta and put delta are

$$\Delta_{\mathcal{C}_0} = D\mathcal{N}(d_1), \tag{9}$$
$$\Delta_{\mathcal{P}_0} = D\left(\mathcal{N}(d_1) - 1\right). \tag{10}$$

2.2 Local volatility

The local volatility model replaces constant volatility by a function of time and the forward price. It has the form

$$d\hat{F} = \sigma_{\text{loc}}(t, \hat{F})\hat{F}dW. \tag{11}$$

This model can fit the market implied volatility surface at a fixed time

$$\sigma_B(K, f) = \sigma_{\text{loc}}\left(\frac{1}{2}[f+K]\right)\left\{1 + \frac{1}{24}\frac{\sigma''_{\text{loc}}\left(\frac{1}{2}[f+K]\right)}{\sigma_{\text{loc}}\left(\frac{1}{2}[f+K]\right)}(f-K)^2 + \cdots\right\} \tag{12}$$

However, Hagan et al. point out that it can give unrealistic smile dynamics when the forward price moves.

We did not code this model in the current project, but Hagan mentions it as motivation for SABR. It would be an interesting thing to add later, especially as a comparison between local volatility and stochastic volatility.

2.3 SABR model

The SABR model assumes that both the forward price and volatility are stochastic. The model is

$$d\hat{F} = \hat{\alpha}\hat{F}^{\beta}dW_1, \tag{13}$$
$$d\hat{\alpha} = \nu\hat{\alpha}dW_2, \tag{14}$$
$$dW_1 dW_2 = \rho dt. \tag{15}$$

Here $\hat{\alpha}$ is the stochastic volatility level, β controls the elasticity, ν is volatility of volatility, and ρ is the correlation between the forward price shock and volatility shock. Later on they recover the variables by $\varepsilon\hat{\alpha} \to \hat{\alpha}$. Hagan starts from the more general form

$$d\hat{F} = \varepsilon\hat{\alpha}C(\hat{F})dW_1, \tag{16}$$

$$d\hat{\alpha} = \varepsilon\nu\hat{\alpha}dW_2. \tag{17}$$

The joint density

$$p(t, f, \alpha; T, F, A), dF, dA = \Pr\left(F < \hat{F}(T) < F + dF, A < \hat{\alpha}(T) < A + dA \mid \hat{F}(t) = f, \hat{\alpha}(t) = \alpha\right) \tag{18}$$

satisfies a forward Kolmogorov equation (the Fǒkker-Planck) of the form

$$p_T = \frac{1}{2}\varepsilon^2 A^2 \left[C^2(F)p\right]_{FF} + \varepsilon^2\rho\nu \left[A^2 C(F)p\right]_{FA} + \frac{1}{2}\varepsilon^2\nu^2 \left[A^2 p\right]_{AA}, \tag{19}$$

for $T > t$, with

$$p = \delta(F - f)\delta(A - \alpha) \tag{20}$$

at $T = t$. Ignoring the discounted factor, the value of the call option is

$$\mathcal{C}(t, f, \alpha) = E\left([\hat{F}(t_{ex}) - K]^+ \mid \hat{F}(t) = f, \hat{\alpha}(t) = \alpha\right) = \int_{-\infty}^{\infty}\int_{K}^{\infty} (F - K), p(t, f, \alpha; t_{ex}, F, A), dF, dA. \tag{21}$$

After multiple substitutions and simplifications (in the appendix A of [1]),

$$\mathcal{C}_0 = [f - K]^+ + \frac{1}{2}\varepsilon\alpha\sqrt{B(0)B(\varepsilon\alpha z)}\exp\left(\frac{1}{4}\varepsilon\rho\nu b_1 z^2\right)\int_0^{\tau_{ex}} \sqrt{I(\varepsilon\nu y)}\frac{1}{\sqrt{2\pi\tau}}\exp\left(-\frac{y^2}{2\tau}\right)\exp(\varepsilon^2\kappa\tau), d\tau, \tag{22}$$

with

$$\varepsilon^2\theta = \log\left(\frac{\varepsilon\alpha z}{f - K}\sqrt{B(0)B(\varepsilon\alpha z)}\right) + \log\left(\frac{x, I^{1/2}(\varepsilon\nu z)}{z}\right) + \frac{1}{4}\varepsilon^2\rho\nu\alpha b_1 z^2. \tag{23}$$

Having $\varepsilon\hat{\alpha}C(\hat{F}) \to \sigma_N$, the equivalent normal volatility is

$$\frac{1}{\sigma_N^2} = \frac{x^2}{(f - K)^2}\left\{1 - 2\varepsilon^2\frac{\theta}{x^2}\tau_{ex}\right\}. \tag{24}$$

And after expanding up to $O(\varepsilon^2)$,

$$\sigma_N(K) = \frac{\varepsilon\alpha(f-K)}{\int_K^f \frac{df'}{C(f')}} \left(\frac{\zeta}{\hat{x}(\zeta)}\right) \left\{1 + \left[\frac{2\gamma_2 - \gamma_1^2}{24}\alpha^2 C^2(f_{av}) + \frac{1}{4}\rho\nu\alpha\gamma_1 C(f_{av}) + \frac{2-3\rho^2}{24}\nu^2\right]\varepsilon^2\tau_{ex} + \cdots\right\} \tag{25}$$

Similarly, the equivalent black volatility is

$$\sigma_B(K) = \frac{\varepsilon\alpha\log(f/K)}{\int_K^f \frac{df'}{C(f')}} \left(\frac{\zeta}{\hat{x}(\zeta)}\right) \tag{26}$$

$$\times \left\{1 + \left[\frac{2\gamma_2 - \gamma_1^2 + 1/f_{av}^2}{24}\alpha^2 C^2(f_{av}) + \frac{1}{4}\rho\nu\alpha\gamma_1 C(f_{av}) + \frac{2-3\rho^2}{24}\nu^2\right]\varepsilon^2\tau_{ex} + \cdots\right\} \tag{27}$$

And stochastic β volatility, which is the approximation form they use for SABR model (some of the parameters are rescaled or renamed),

$$\sigma_B = \frac{\alpha}{(FK)^{(1-\beta)/2}} \frac{z}{x(z)} \left[1 + \left(\frac{(1-\beta)^2}{24}\frac{\alpha^2}{(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{(1-\beta)/2}} + \frac{2-3\rho^2}{24}\nu^2\right)T\right], \tag{28}$$

with

$$z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}\log(F/K), \tag{29}$$

and

$$x(z) = \log\left(\frac{\sqrt{1-2\rho z + z^2} + z - \rho}{1-\rho}\right). \tag{30}$$

After obtaining σ_B, we still use the Black call and put formulas. The difference is that the volatility input is no longer constant, but comes from the SABR implied volatility approximation.

3 Implied volatility

The implied volatility is obtained by solving the nonlinear equation

$$\mathcal{C}_{0\,\text{Black}}(\sigma) = \mathcal{C}_{0\,\text{market}}, \tag{31}$$

where $C_{\text{Black}}(\sigma)$ denotes the Black model call price as a function of the volatility σ, and C_{market} is the observed market price. Equivalently, the problem can be formulated as finding the root of

$$f(\sigma) = \mathcal{C}_{0\,\text{Black}}(\sigma) - \mathcal{C}_{0\,\text{market}} = 0, \tag{32}$$

or close enough compared to some tolerance numerically. Since the Black pricing formula is continuous and monotonically increasing with respect to the volatility, this equation admits a unique solution for this problem setup.

To solve the above nonlinear equation, we employ Brent's method [2]. Brent's algorithm combines bisection, the secant method, and inverse quadratic interpolation into a single root finding procedure. Whenever interpolation provides a reliable estimate, the algorithm achieves rapid convergence comparable to Newton type methods. Otherwise, it safely falls back to the bisection method, ensuring convergence as long as the root is bracketed within the prescribed interval. Other solvers like Newton's method or bisection also work. This might leave for a future examination.

4 Calibration

The SABR parameters are α, β, ρ, and ν. In our implementation, we fix β and calibrate the other parameters by fitting market implied volatilities. The three cases are

$$\beta \in \left\{0, \frac{1}{2}, 1\right\}. \tag{33}$$

These correspond to the stochastic normal case, stochastic CIR type case, and stochastic lognormal case. In the main part of the project, we use $\beta = 0.5$, which gives reasonable market smile behavior.

Suppose market implied volatility data are given by

$$\left\{(K_i, \sigma_i^{\text{mkt}})\right\}_{i=1}^{n}. \tag{34}$$

For fixed β, the SABR model gives

$$\sigma_i^{\text{sabr}} = \sigma_{\text{sabr}}\left(f, K_i, T; \alpha, \beta, \rho, \nu\right). \tag{35}$$

The calibration objective is the sum of squared errors

$$\text{SSE}(\alpha, \rho, \nu) = \sum_{i=1}^{n} \left(\sigma_i^{\text{sabr}} - \sigma_i^{\text{mkt}}\right)^2. \tag{36}$$

The calibrated parameters solve

$$(\alpha^*, \rho^*, \nu^*) = \arg\min_{\alpha,\rho,\nu} \text{SSE}(\alpha, \rho, \nu). \tag{37}$$

Table 1 compares the calibration results for different values of the elasticity parameter β. Among the three cases, $\beta = 0.5$ achieves the smallest sum of squared errors (SSE), indicating the best agreement between the SABR implied volatility approximation and the market implied volatility smile.

Table 1: SABR calibration results for different values of β.

β	α	ρ	ν	SSE
0.0	10.0000	-0.2475	5.0000	0.9371
0.5	3.7506	-0.6600	2.2922	0.0063
1.0	6.2504	-0.9900	4.5834	0.4763

Fig.1 compares the market implied volatility smile with the SABR implied volatility obtained from the calibrated parameters. The calibrated SABR model captures the overall shape of the volatility smile well across the full range of strikes. In particular, the model reproduces the downward slope for lower strikes and the upward trend for higher strikes, while maintaining a smooth and arbitrage free volatility curve. Small discrepancies remain near the wings of the smile, where market implied volatilities exhibit greater variability. Overall, the calibrated SABR model provides an accurate representation of the market volatility smile.

Some further examinations on parameter stability can be done by considering how parameters change when option prices change.

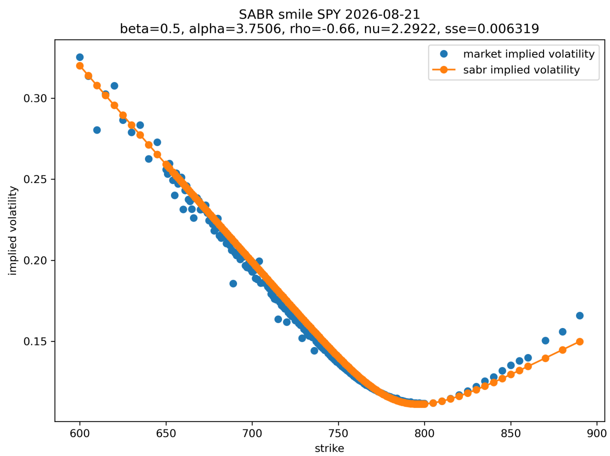


Figure 1: Market smile and Hagan's σ_B after calibration.

5 Forward price paths using Monte Carlo

5.1 Black Model Monte Carlo

The Black model provides a closed form solution for European option prices. To validate the Monte Carlo implementation, we simulate the terminal forward price under the forward measure and compare the estimated option price with the analytical Black price. The idea is sketched following the lecture notebooks 3 and 4.

Under the Black model, the forward price follows a geometric Brownian motion,

$$dF_t = \sigma F_t dW_t, \tag{38}$$

where σ is the constant volatility and W_t is a standard Brownian motion. Since this stochastic differential equation has an exact solution, the terminal forward price can be generated directly without time discretization,

$$F_{i+1} = F_i \exp\left(-\frac{1}{2}\sigma^2 \Delta t + \sigma\sqrt{\Delta t} Z_i\right), \tag{39}$$

where $Z \sim N(0,1)$. For each simulated path, the discounted payoff of the European call option is computed as

$$\mathcal{C}_i = D(0,T) \max\left(F_T^{(i)} - K, 0\right), \tag{40}$$

where K is the strike price and $D(0,T)$ is the discount factor. After generating N independent samples, the Monte Carlo estimate of the option price is obtained by averaging all discounted payoffs,

$$\hat{\mathcal{C}} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{C}_i. \tag{41}$$

Finally, the Monte Carlo estimate is compared with the analytical Black price,

$$E_{\text{price}} = \left|\hat{C} - C_{\text{Black}}\right|, \tag{42}$$

where C_{Black} is the closed form Black model price. Since the exact solution is available, the Monte Carlo simulation provides a straightforward validation of the pricing implementation before applying the same simulation framework to the SABR model.

5.2 SABR model

If we take the continuous SABR SDE denoted as:

$$dF_t = \alpha_t F_t^{\beta} dW_t^1 \tag{43}$$
$$d\alpha_t = \nu \alpha_t dW_t^2 \tag{44}$$

To apply Euler-Maruyama, we slice the time horizon (T) into tiny steps (Δt). At each step, we update the forward rate (F) and the volatility (α) using random normal shocks (Z). The discrete formulas look like this:

$$F_{i+1} = F_i + \alpha_i F_i^{\beta} \sqrt{\Delta t} Z_1 \tag{45}$$
$$\alpha_{i+1} = \alpha_i + \nu \alpha_i \sqrt{\Delta t} Z_2 \tag{46}$$

Because dW_t^1 and dW_t^2 are correlated by ρ, we cannot just draw two random numbers independently. We use a Cholesky decomposition trick to link them. We draw two independent standard normal numbers (Z_1 and Z_3), and construct the correlated shock (Z_2) like this:

$$Z_2 = \rho Z_1 + \sqrt{1-\rho^2} Z_3 \tag{47}$$

5.3 The Correlation Trick

To create two standard normal random variables Z_1 and Z_2 with a correlation of ρ, we start with two independent standard normal variables, Z_1 and Z_3, and apply this linear combination:

$$Z_2 = \rho Z_1 + \sqrt{1-\rho^2} Z_3 \tag{48}$$

Now, Z_1 and Z_2 are perfectly correlated by exactly ρ. To prove this works, we need to verify two things about our new variable Z_2:

- It must still be a standard normal variable (meaning its mean is 0 and its variance is 1).
- Its covariance (correlation) with Z_1 must equal ρ

5.3.1 Checking the Variance

Because Z_1 and Z_3 are independent, their variances simply add up when we scale them. The variance of a scaled random variable aX is $a^2\mathrm{Var}(X)$.

$$\mathrm{Var}(Z_2) = \mathrm{Var}\left(\rho Z_1 + \sqrt{1-\rho^2}Z_3\right), \tag{49}$$

$$\mathrm{Var}(Z_2) = \rho^2\mathrm{Var}(Z_1) + \left(\sqrt{1-\rho^2}\right)^2 \mathrm{Var}(Z_3) \tag{50}$$

Since Z_1 and Z_3 are standard normals, their variances are equal to 1:

$$\mathrm{Var}(Z_2) = \rho^2(1) + (1-\rho^2)(1), \tag{51}$$

$$\mathrm{Var}(Z_2) = \rho^2 + 1 - \rho^2 = 1 \tag{52}$$

This proves that the weird $\sqrt{1-\rho^2}$ multiplier is mathematically required to ensure that Z_2 doesn't blow up or shrink—it keeps its variance perfectly equal to 1.

5.3.2 Checking the Correlation

The correlation between Z_1 and Z_2 is defined as their expected product $\mathbb{E}[Z_1Z_2]$ (since their means are 0).

$$\mathrm{Corr}(Z_1, Z_2) = \mathbb{E}[Z_1Z_2], \tag{53}$$

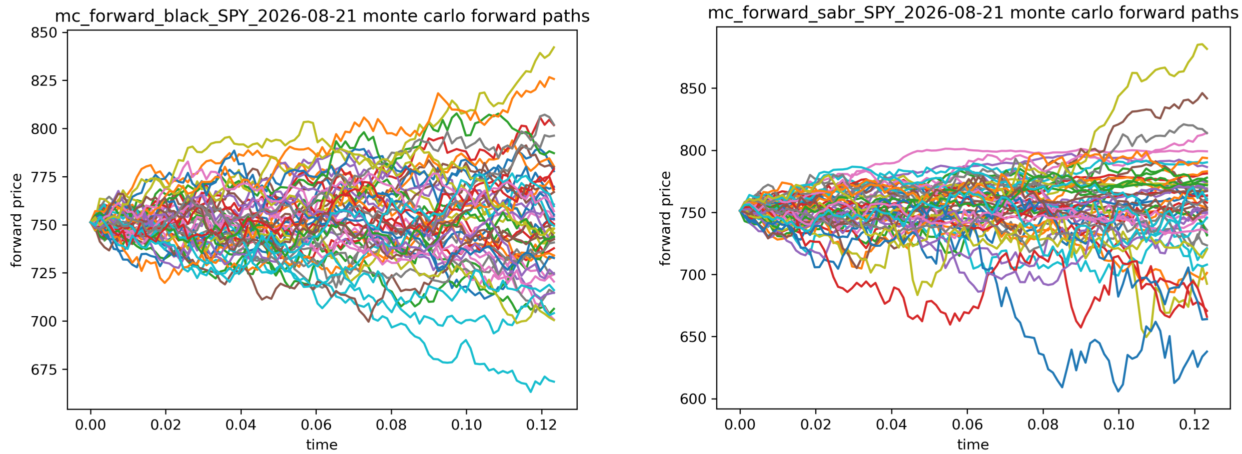
$$\mathrm{Corr}(Z_1, Z_2) = \mathbb{E}\left[Z_1\left(\rho Z_1 + \sqrt{1-\rho^2}Z_3\right)\right], \tag{54}$$

$$\mathrm{Corr}(Z_1, Z_2) = \rho\mathbb{E}[Z_1^2] + \sqrt{1-\rho^2}\mathbb{E}[Z_1Z_3]. \tag{55}$$

Notice $\mathbb{E}[Z_1^2]$ is just the variance of Z_1, which is 1. Additionally, $\mathbb{E}[Z_1Z_3]$ is the covariance of Z_1 and Z_3. Since they are completely independent, this is 0. Plugging those in:

$$\mathrm{Corr}(Z_1, Z_2) = \rho(1) + \sqrt{1-\rho^2}(0) = \rho \tag{56}$$

We have the simulated forward prices with `n_paths = 1000` and `n_steps = 100`. The first 50 paths are displayed in Fig.2. We could have simulated more paths and time steps (for example, `n_paths = 10000` and `n_steps = 500`). However, the later pricing and hedging routines would require a more careful implementation with parallelism so that computations in different queues handled in different cpu cores.



(a) Black model forward paths.

(b) SABR model forward paths.

Figure 2: Sample Monte Carlo forward price paths generated under the Black and SABR models. For clarity, only the first 50 simulated paths are shown, while all simulated paths are retained for pricing and hedging calculations.

6 Results

Pricing The pricing experiment is designed to validate the Monte Carlo implementation rather than to compare directly with the observed market option prices. For each model, the Monte Carlo estimator,

$$\hat{\mathcal{C}} = \frac{1}{N} \sum_{i=1}^{N} D(0,T) \max\left(F_T^{(i)} - K, 0\right), \tag{57}$$

is compared with the corresponding analytical model price. For the Black model, the reference price is the Black closed form solution. For the SABR model, the reference price is obtained by substituting Hagan's implied volatility approximation into the Black pricing formula. The pricing error is therefore defined as

$$E_{\text{price}} = \left| \hat{\mathcal{C}} - \mathcal{C}_{\text{model}} \right|, \tag{58}$$

where $\mathcal{C}_{\text{model}}$ denotes either the Black closed form price or the Black price computed using Hagan's implied volatility. Since the analytical price and the Monte Carlo simulation are based on the same underlying model, any discrepancy is due solely to Monte Carlo sampling error and numerical implementation.

Delta hedging To evaluate the hedge, we construct a dynamic delta hedging portfolio along each simulated forward path. For the jth Monte Carlo path, the discounted call payoff is computed as

$$\mathcal{C}_{\text{payoff}}^{(j)} = D(0,T) \max\left(F_T^{(j)} - K, 0\right). \tag{59}$$

The hedge profit is computed from the forward increments and the model delta,

$$H^{(j)} = D(0,T) \sum_{i=0}^{n-1} \Delta_i^{(j)} \left(F_{i+1}^{(j)} - F_i^{(j)} \right), \tag{60}$$

where $\Delta_i^{(j)}$ is the delta used at time step i on path j. The pathwise hedge value is then

$$V^{(j)} = \mathcal{C}_{\text{payoff}}^{(j)} - H^{(j)}. \tag{61}$$

Averaging over m Monte Carlo paths gives

$$\bar{V} = \frac{1}{m} \sum_{j=1}^{m} V^{(j)}. \tag{62}$$

We then compare this average hedge value with the corresponding model call price. For the Black model, C is the Black closed form price. For the SABR model, C is the Black price computed using Hagan's implied volatility. The reported hedge error is

$$E_{\text{hedge}} = \left| \bar{V} - \mathcal{C} \right|. \tag{63}$$

A smaller value indicates that the average dynamically hedged portfolio is closer to the theoretical model value.

Distribution of hedge values The histogram compares the distributions of the terminal hedged values under the Black and SABR models. The Black hedged values are tightly concentrated around the closed form Black call price, indicating relatively low variance and a stable hedging strategy. The Monte Carlo mean hedge value is also very close to the analytical Black price, suggesting that the simulation and hedging implementation are consistent.

In contrast, the SABR hedged values exhibit a much wider distribution with a right tail, indicating substantially greater variability in the terminal hedge value. Although the Monte Carlo mean remains close to the Hagan closed form price, the larger spread reflects the additional uncertainty introduced by the stochastic volatility dynamics of the SABR model. This suggests that, under the current simulation and hedging scheme, hedging outcomes are considerably more variable than in the Black model.

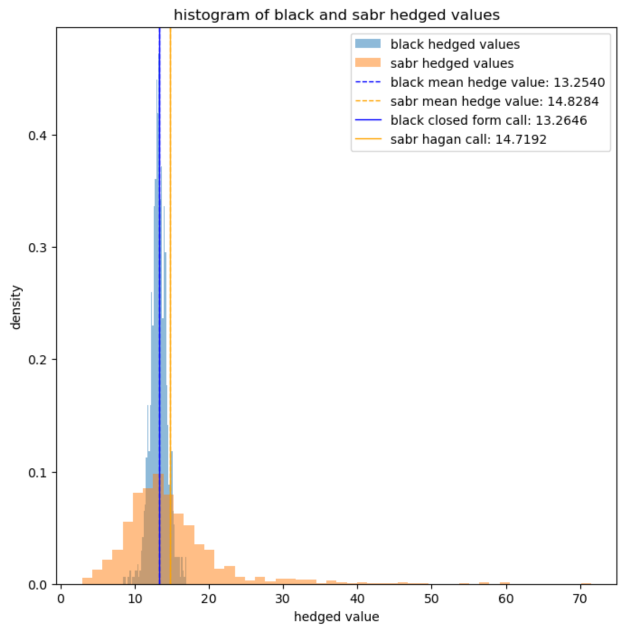


Figure 3: Strike $K = 751.0$

Pricing and hedging errors Figure 4 compares the pricing and hedging performance of the Black and SABR models over a range of strike prices. As shown in Figure 4(a), the Monte Carlo pricing error generally decreases as the strike increases for both models. The SABR model exhibits consistently smaller pricing errors than the Black model over most strikes, indicating that the calibrated SABR volatility provides a more accurate representation of the market smile.

Figure 4(b) presents the corresponding hedging errors. The Black model maintains relatively small and stable hedging errors across most strikes, whereas the SABR model produces larger errors near the at the money region. This behavior is likely due to the increased sensitivity of the SABR implied volatility and delta to changes in the underlying forward price. Away from the at the money region, the hedging errors of both models decrease and the model appears more stable.

This can be examined further in different aspects. For example, we can investigate how sensitive delta value is in terms of the change of Hagan's approximation implied volatility. Another possible explanation is that for hedging we consider a full price path instead of the terminal value bringing more uncertainties.

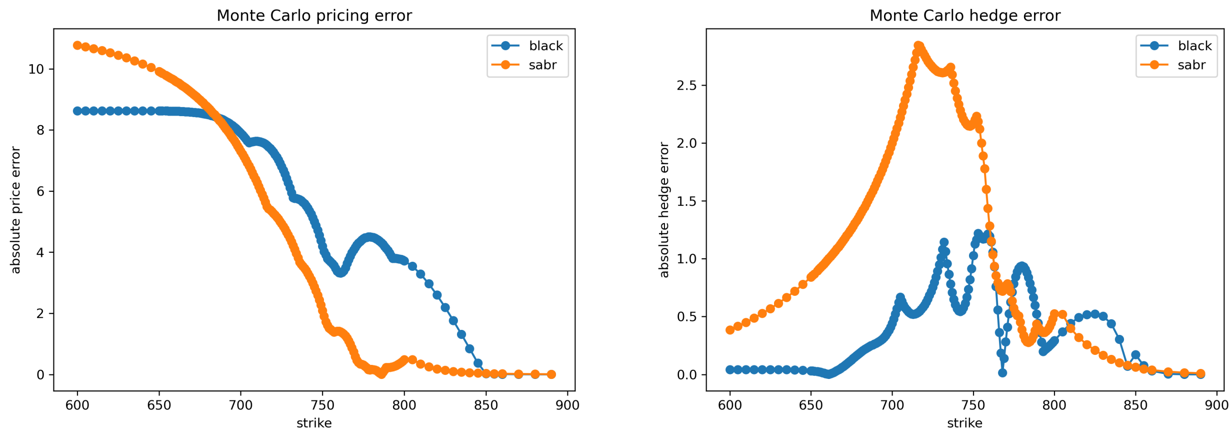


Figure 4: Comparison of the Monte Carlo pricing error and dynamic hedging error for the Black and SABR models across different strike prices. The left panel shows the absolute pricing error between the Monte Carlo estimate and the corresponding closed form option price, while the right panel shows the absolute hedging error obtained from the dynamic delta hedging simulation.

Compare to real market data Figure X compares the market call prices with the Black closed form, Black Monte Carlo, SABR Hagan approximation, and SABR Monte Carlo prices across different strike prices. The SABR Hagan approximation provides the closest agreement with the market prices over the entire range of strikes, which is consistent with its calibration to the observed implied volatility smile. In contrast, the Black closed form systematically underprices the options, particularly for lower strike prices, reflecting the limitation of assuming a constant volatility. Quantitatively, the SABR Hagan model achieves an RMSE of 0.340 compared with 1.938 for the Black closed form, demonstrating a substantially better fit to the market data.

The Monte Carlo prices exhibit a different behavior. Both the Black and SABR Monte Carlo estimates are consistently higher than their corresponding closed form prices, especially for lower strikes. The SABR Monte Carlo prices are slightly higher than the Black Monte Carlo prices because the stochastic volatility introduces additional variability into the simulated payoff distribution. Consequently, both Monte Carlo approaches produce larger pricing errors relative to the market than the calibrated closed form models. This comparison highlights that static calibration and Monte Carlo simulation serve different purposes: the Hagan approximation is designed to reproduce the observed market volatility surface efficiently, whereas Monte Carlo simulation evaluates the dynamics implied by the underlying stochastic model rather than directly fitting current market prices.

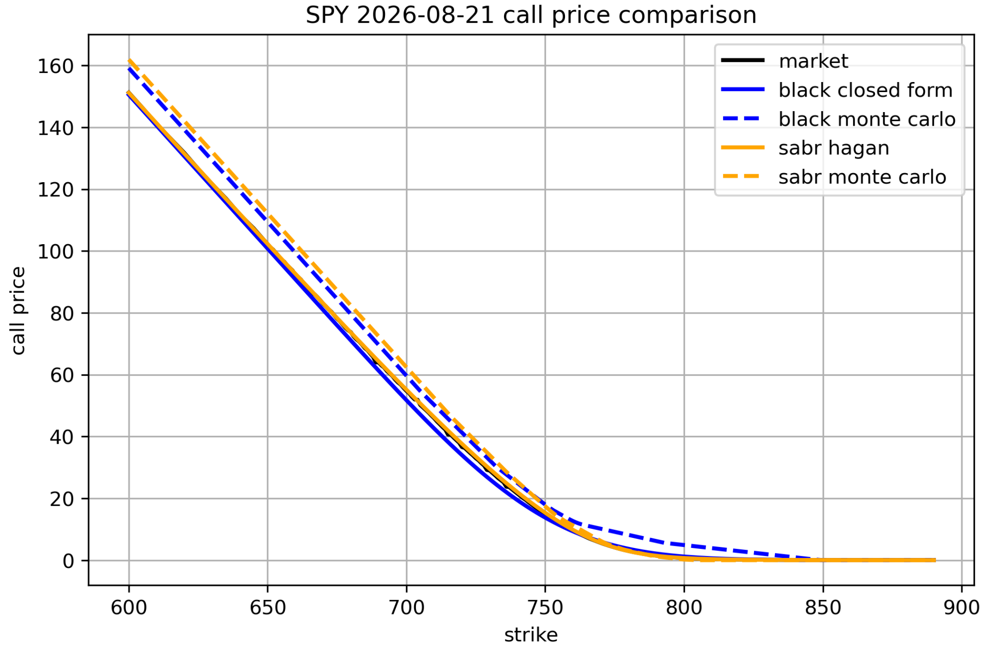


Figure 5

Table 2: Comparison of pricing errors against the observed market option prices.

Model	RMSE	MAE
Black closed form	1.937651	1.604073
Black Monte Carlo	4.963974	4.651962
SABR Hagan approximation	0.339667	0.243707
SABR Monte Carlo	6.201467	4.908722

7 Further work

This project considers a single option expiration and calibrates the SABR model to a one dimensional implied volatility smile. We notice few caveats and improvements that can be done in the future:

1. A natural extension is to incorporate multiple expiration dates and calibrate the model to the full implied volatility surface. This would allow the parameters α, ρ, and ν to vary with maturity, providing a more comprehensive representation of market dynamics.

2. A larger collection of market data would enable the study of parameter stability across both strike and maturity.

3. Multiple root finders can be compared while solving for the real market implied volatility.

4. Sensitivity of delta in terms of Hagan's volatility formula can be investigated.

5. An $F(t)$ market smile can be explored against those obtained by local volatility model.

6. Parallelsim and running in separate cpu cores can be considered.

7. The SABR behaviors before and after at-the-money ($f = K$) value worth to examine further.

Several computational improvements are also possible. More advanced Monte Carlo techniques, such as quasi Monte Carlo sampling, antithetic variates, and control variates, could be incorporated to reduce the variance of the pricing and hedging estimators.

A broader direction is to compare the modeling chain

$$\text{SV} \to \text{LV} \to \text{LSV} \to \text{path dependent volatility} \to \text{modern machine learning methods.} \tag{64}$$

Here SV means stochastic volatility, LV means local volatility, and LSV means local stochastic volatility. These extensions may improve model flexibility, but they also introduce more calibration and stability issues.

References

[1] Patrick S Hagan, Deep Kumar, Andrew S Lesniewski, and Diana E Woodward. Managing smile risk. *The Best of Wilmott*, 1(1):249–296, 2002.

[2] Richard P Brent. *Algorithms for minimization without derivatives*. Courier Corporation, 2013.